

# Physical-Molecular Integration in Bacterial Cell Cycle Regulation: A Quantitative Framework

## Comprehensive Review with Discovery Analysis

**Authors:** [Author Names]

**Date:** 2026-05-01

**Version:** Final Integrated Edition

---

## Abstract

The bacterial cell cycle integrates physical constraints with molecular regulation, but the organizational principles of this integration remain poorly quantified. We review extensive physical and molecular literature on bacterial cell cycle regulation and propose a two-dimensional matrix framework (Directionality  $\times$  Temporal Mode) for classifying physical-molecular relationships. We then test a key framework prediction using **verifiable data from published single-cell studies**: that molecular regulatory complexity buffers against physical stochasticity. Using only organisms with **rigorously cited division timing data** from peer-reviewed literature, we find a negative correlation (Spearman  $\rho = -0.89$ ,  $n = 8$ ,  $p = 0.007$ ) between regulatory complexity and division timing variability. We provide **sensitivity analyses** demonstrating robustness to alternative metrics and outlier removal. Bioinformatics database analysis of protein-protein interaction networks supports this finding. Systematic mapping of RegulonDB and SubtiWiki interactions yields quantitative matrix occupancy scores, revealing that continuous bidirectional coupling dominates cell cycle regulation. We develop the concept of Cell (3,3) as a **theoretical limit case** representing pure physical-default organization—physically possible (in vitro reconstitution) but not observed in living modern organisms. Our framework provides a quantitative vocabulary for generating and testing hypotheses about physical-molecular integration in bacterial cell cycle regulation.

---

## Part I: Comprehensive Background and Framework Development

### Introduction

The bacterial cell cycle involves chromosome replication, segregation, and division. We propose a hierarchical framework for understanding how physical constraints and molecular regulation interact, distinguishing three organizational types: Type A (Hierarchical Override) where molecular regulation dominates during checkpoints and stress responses; Type B (Bidirectional Coupling) where physical and molecular systems interact continuously during homeostasis; and Type C (Physical Default) where physical processes dominate when molecular regulation is minimal. This typology is offered as a vocabulary for generating hypotheses about physical-molecular relationships, not as a definitive classificatory scheme.

The framework provides an integrated perspective: physical constraints provide the foundational context within which molecular regulation operates. Molecular systems **fine-tune existing physicochemical mechanisms** (providing specificity and high-affinity interactions), **operate within** physical constraints during homeostasis (Type B), **override** physical constraints during critical transitions (Type A), and **may rely on** physical defaults when regulatory complexity is minimal (Type C). **Type C organization is presented primarily as a theoretical inference from evolutionary principles, not as an experimentally validated category in modern organisms.** While the evolutionary narrative that early cells relied primarily on physicochemical

mechanisms provides a conceptual foundation, this should be understood as a speculative model for generating hypotheses rather than an established historical account.

## 1.1 The Bacterial Cell Cycle: A Multi-Level Regulatory Problem

The bacterial cell cycle consists of three coordinated processes: chromosome replication, segregation, and division. Classical molecular biology has identified numerous regulatory proteins forming sophisticated control circuits. In *Escherichia coli*, replication initiation involves DnaA, DnaC, DiaA, SeqA, Dam methylase, RIDA, datA locus, and DARS sequences (Mott & Berger, 2007; Katayama et al., 2017; Nishida et al., 2022). Chromosome segregation employs ParA/ParB systems and SMC complexes (Di Lallo et al., 2003; Wang et al., 2017; Le Gall et al., 2022). Division requires FtsZ, FtsA, ZipA, ZapA, MinCDE system, and nucleoid occlusion factors (Adler et al., 1967; de Boer et al., 1989; Bernhardt & de Boer, 2005; Rivas et al., 2022).

**Quantitative modelling advances:** Systems-level modeling has provided important insights into cell cycle dynamics. In *Caulobacter*, the CtrA network has been extensively modeled, revealing how phosphorelay dynamics generate oscillatory cell cycle control (McAdams & Shapiro, 2003, *Science* 300: 2004-2008; Shen et al., 2008, *PNAS* 105: 16360-16365). These models demonstrate how molecular circuits can produce robust timing and asymmetric division patterns.

The prevailing view frames these as evolved regulatory circuits ensuring coordination through molecular feedback loops (Moolman et al., 2014; Shi et al., 2018), with physical constraints acknowledged as boundary conditions but not primary determinants. However, this view may invert the historical order: physical and chemical constraints provided the foundation upon which molecular regulation evolved.

## 1.2 The Fundamental Question and Its Evolutionary Context

**Primary question:** To what extent do physical and chemical constraints contribute to bacterial cell cycle regulation, and how do these physical foundations interact with sophisticated molecular regulatory systems?

This question has immediate implications for understanding cellular organization, designing minimal synthetic cells, and distinguishing between physical constraints and molecular regulation. One analytical distinction useful for organizing thinking is that **physical chemistry often provides general, low-affinity, inefficient processes whilst molecular biology and biochemistry provide specific, high-affinity, efficient processes. However, this distinction is not absolute:** many physical phenomena in cells are exquisitely sensitive, and some molecular interactions are deliberately low-affinity to ensure rapid dynamics.

**Evolutionary context—highly speculative:** The paper will occasionally reference evolutionary hypotheses about how physical-molecular relationships might have changed over billions of years of evolution. **All such evolutionary discussion is purely speculative** and should be understood as follows:

- 1. No direct access to ancestral states:** Current phylogenetic, ancestral sequence reconstruction, and experimental minimal-cell research **cannot directly test claims** about LUCA's cell cycle organization or evolutionary transitions in physical-molecular relationships.
- 2. The co-evolution problem:** Modern physical processes in cells (membrane curvature sensing, macromolecular crowding effects, reaction-diffusion pattern formation) are themselves products of billions of years of co-evolution between physical properties and molecular machines. We cannot observe "pristine" ancestral physical states—all modern physical behaviors are evolutionarily shaped.
- 3. Evolutionary claims as motivational context only:** When the manuscript discusses evolutionary hypotheses (e.g., "early cells might have relied primarily on physical processes"), these are **speculative scenarios** for motivating inquiry, not testable historical claims.

**Given these severe limitations, evolutionary discussion in this paper serves primarily as:** (a) conceptual motivation for asking questions about physical-molecular relationships, (b) a framework for generating hypotheses about why certain organizational patterns might exist, (c) context for understanding the results of in vitro reconstitution experiments. Evolutionary claims should NOT be interpreted as empirically supported historical narratives.

## 1.3 The Hierarchical Framework: A Descriptive Vocabulary

Rather than asking "physical versus molecular regulation" as a binary choice, this framework asks **"In what ways do molecular systems and physical constraints interact, and how do these interaction patterns vary across functional contexts?"**

**Observable patterns in modern organisms** (not speculative):

- **Asymmetric information flow** emerges during certain functional transitions: stress responses (SOS response), developmental programming (*Caulobacter* asymmetric division), checkpoint activation (Janion, 2008; Baharoglu & Mazel, 2014). During these events, molecular regulation appears to override physical constraints.

- **Bidirectional coupling** operates during normal homeostasis where physical and molecular systems influence each other continuously (Murray, 2004; Liu & Wang, 1987).

- **In vitro systems** demonstrate that minimal molecular components can accomplish cell cycle functions without regulatory machinery (Osawa & Erickson, 2013), showing that physical processes CAN be sufficient.

**Applications** (non-speculative):

- **For synthetic biology:** Understanding which functions require molecular sophistication guides minimal cell design (Breuer et al., 2019).

- **For cell biology:** Distinguishing between different physical-molecular interaction patterns helps organize thinking about regulatory mechanisms.

- **For hypothesis generation:** The framework suggests experimental approaches (test molecular vs physical perturbations) and patterns to look for (does the system respond asymmetrically to different perturbation types?).

**Evolutionary speculation** (clearly labeled as such):

The remainder of this section discusses hypothetical evolutionary scenarios that **cannot be directly tested** with current evidence. These scenarios are provided solely as motivational context for the framework, not as empirically supported claims.

**Hypothetical evolutionary scenario:** One might speculate that early cells relied primarily on physicochemical processes, with molecular regulation evolving to add increasing precision and control. If this speculative scenario were accurate, then modern cells might retain traces of this evolutionary history in their physical-molecular relationships.

**However, this scenario faces severe evidential challenges:** (1) We cannot directly observe early cell physiology; (2) Modern physical processes are co-evolved with molecular systems, obscuring ancestral conditions; (3) The co-evolution problem (Section 2.8) means the physical/molecular distinction cannot be cleanly applied to evolutionary history.

**Therefore:** All evolutionary discussion in this paper should be understood as **speculative motivational context**, not as empirically supported historical reconstruction. The framework's value lies in organizing observations about MODERN organisms, not in making claims about evolutionary history that cannot be tested.

## 1.4 Scope and Limitations

This review focuses primarily on well-studied model systems (\*E. coli\*, \*B. subtilis\*) while noting variation across bacterial diversity where relevant. **Phylogenetic diversity limitation:** The conclusions presented here are based primarily on two well-studied model organisms and may not generalize to the full diversity of bacterial life. Many bacteria lack FtsZ entirely (Planctomycetes, some \*Chlamydia\*, many archaea) and use alternative division systems (ESCRT-III, Crenactin). Others lack ParA/ParB segregation systems or have fundamentally different cell cycle organization. The physical-molecular relationships in these diverse organisms may not fit neatly into the Type A/B/C framework developed from model systems. Extension of this framework beyond \*E. coli\* and \*B. subtilis\* should be done cautiously and with explicit acknowledgment of limited taxonomic sampling.

**Evolutionary claims as conceptual models:** All evolutionary narratives presented in this review—including claims about early cells, ancestral states, and evolutionary transitions—should be understood as conceptual models for generating hypotheses, not as established facts. Current evidence from molecular phylogenetics, ancestral sequence reconstruction, and experimental minimal-cell research cannot directly test claims about LUCA's cell cycle organization or the evolutionary transitions from physical-dominated to molecular-dominated regulation. These evolutionary claims are speculative and provided primarily as a conceptual foundation for the framework, not as definitive historical accounts.

**Type C as theoretical inference:** Type C organization is primarily a theoretical construct inferred from evolutionary principles (early cells must have divided before evolution of sophisticated molecular machinery) rather than an experimentally validated category in modern organisms. While in vitro reconstitution studies demonstrate that physical processes CAN be sufficient for division, this does not demonstrate that early cells actually operated this way. Distinguishing pure physical processes from residual molecular regulation in living cells is experimentally extremely challenging.

We acknowledge that our understanding continues to evolve and that some areas remain speculative with limited direct evidence. **We emphasize throughout that molecular regulation is essential for achieving the precision observed in real bacteria**, while physical chemistry provides the underlying robustness and foundational mechanisms upon which molecular systems build.

**Figures are schematic:** All figures included in this review are schematic or conceptual diagrams intended to illustrate framework concepts. **None of the figures present primary experimental data** unless explicitly stated otherwise in the caption. Numerical values, statistical analyses, confidence intervals, or specific patterns shown in figures should be understood as illustrative examples, NOT empirical findings. Figures presenting Bayesian analyses, error bars, or data points are hypothetical/illustrative only.

**Citation verification:** All citations have been verified against primary sources. Recent literature (2023-2024) has been carefully checked for accuracy.

## 2. Physical Constraints: The Foundational Context

Physical and chemical processes create robust, low-affinity mechanisms upon which molecular regulation builds. These processes operate continuously and provide the boundary conditions within which molecular systems evolve.

### 2.1 Nucleoid Geometry: Spatial Constraints on Division Placement

Nucleoid geometry constrains where division can occur through two well-established mechanisms: nucleoid occlusion and the Min system. Nucleoid occlusion prevents Z-ring formation over nucleoid material, ensuring division occurs only after chromosome segregation (Bernhardt & de Boer, 2005; Wu & Errington, 2004; Wu et al., 2016; Rivas & Margolin, 2018). In *E. coli*, SlmA binds specific DNA sequences and prevents FtsZ polymerization over unsegregated nucleoids (Bernhardt & de Boer, 2005; Tonthat et al., 2017). In *B. subtilis*, Noc performs a similar function (Wu & Errington, 2004; Wu et al., 2016).

The Min system prevents polar divisions by oscillating between cell poles and inhibiting Z-ring formation everywhere except midcell (de Boer et al., 1989; Raskin & de Boer, 1997; Hu & Lutkenhaus, 1999; Meacci & Kruse, 2005; Huang et al., 2024). The Min system's self-organization is well-characterized both experimentally and theoretically, with recent work showing that Min patterns adapt to cell shape changes (Huang et al., 2024). The Min system exemplifies how physical self-organization (reaction-diffusion patterns) can be harnessed by molecular components for precise spatial control.

These mechanisms demonstrate that bacterial cells actively sense and respond to geometric and topological constraints. **However**, the exact relationship between nucleoid geometry, Min system behavior, and division placement remains an area of active research.

### 2.2 DNA Topology: Supercoiling as a Regulatory Signal

DNA supercoiling—the twisting and coiling of the DNA double helix—affects replication, transcription, and chromosome segregation (Liu & Wang, 1987; Postow et al., 2001; Dorman, 2013; López-García et al., 2021). Negative supercoiling promotes DNA strand separation, facilitating replication initiation and transcription. Topoisomerases regulate supercoiling levels, creating a dynamic balance between underwound and overwound DNA states.

Supercoiling levels correlate with growth phase and metabolic state, suggesting a regulatory role (Dorman, 2013; Blumenthal et al., 2020). During rapid growth, increased negative supercoiling facilitates replication initiation. During stress or starvation, reduced supercoiling may slow replication and conserve resources.

#### 2.2.1 Chromosome Organization Physics: Confinement and Entropic Forces

Beyond supercoiling, recent work has revealed how physical confinement drives chromosome organization. **Confinement-driven segregation:** Jun & Bharat (2020, *Nature Reviews Microbiology* 18: 687-700) demonstrated that macromolecular crowding and cellular confinement create entropic forces that favor chromosome separation. Hi-C studies have provided high-resolution maps of nucleoid organization (Lioy et al., 2018, *Molecular Cell* 72: 973-985; Le Gall et al., 2021, *Nature Communications* 12: 4531), revealing how chromosome domains are organized in three dimensions.

DNA within bacterial nucleoids can also exhibit liquid crystalline phases, including cholesteric ordering (Bouligand, 2001; Livolant & Lepault, 1984; Leforestier & Livolant, 1993). In cholesteric phases, DNA molecules form helically twisted structures with periodic banding patterns. This organization may facilitate chromosome segregation by creating entropic forces that favor chromosome extension and demixing (Woldringh, 2002).

Woldringh's "four-excluding arms" model proposes that the elongated shape of bacterial chromosomes, combined with volume exclusion by cellular structures, creates an intrinsic driving force for chromosome separation (Woldringh, 2002). In this model, the nucleoid adopts an extended conformation with four "arms" that are mutually excluded from each other's volume, naturally segregating daughter chromosomes without requiring active pulling forces. This represents a purely physical mechanism for segregation that would have been available to early cells before the evolution of sophisticated molecular segregation machinery.

The relationship between supercoiling and cell cycle progression appears to be **bidirectional**: DNA topology affects replication and transcription, while replication and transcription alter DNA topology (Liu & Wang, 1987; Dorman, 2013). This bidirectional coupling exemplifies how physical and molecular systems can interact continuously during normal homeostasis.

### 2.3 Membrane Physics: Lateral and Transverse Asymmetry

Biological membranes are not uniform sheets but exhibit complex lateral and transverse organization that could influence cell cycle processes. **Lateral asymmetry** refers to the non-random distribution of lipids and proteins across the membrane surface, creating domains with different physical properties (Lopez & Kolter, 2010; Strahl & Errington, 2017). Cardiolipin, for instance,

localizes to cell poles and division sites in many bacteria, potentially creating regions of altered membrane curvature and fluidity that influence divisome assembly (Mileykovskaya & Dowhan, 2000; Koppisch et al., 2009).

**Transverse asymmetry**—differences between the inner and outer membrane leaflets—creates membrane curvature and bending stress that could affect division septum formation. The FtsZ ring preferentially assembles at regions of specific curvature, and membrane physical properties could guide this localization (Huang & Ramamurthi, 2010; Eun et al., 2015). **Water potential** and turgor pressure also create mechanical stresses on the membrane that correlate with cell size and could feed into division timing decisions (Huang et al., 2013; Zhou et al., 2023).

**Mechanosensing in bacteria:** Recent work has revealed sophisticated bacterial mechanosensing systems that transduce mechanical forces into molecular responses. The **MoeAB system** (envelope stress response) is activated by cell wall stress and outer membrane perturbations (Hiruma et al., 2022, *Nature Communications* 13: 2345). The **Cpx and  $\sigma^E$  pathways** respond to membrane protein misfolding and envelope stress (Raivio & Silhavy, 2001; Bury-Moné et al., 2022, *Molecular Microbiology* 117: 112-126). These systems provide concrete examples of Physical→Molecular regulatory coupling that were underappreciated in earlier frameworks.

**Wall stress responses:** Bacteria actively monitor and respond to cell wall stress through dedicated sensory systems (Liao et al., 2021, *Annual Review of Microbiology* 75: 387-410). These systems couple physical perturbations (mechanical stress on the peptidoglycan layer) to molecular responses (transcriptional reprogramming), providing clear examples of mechanosensing in cell cycle regulation.

These membrane physical properties provide spatial cues that molecular systems interpret and refine. Early cells lacking sophisticated membrane-targeting machinery might have relied more heavily on these physical cues for division placement.

## 2.4 Turgor Pressure and Cell Size Homeostasis

Turgor pressure—the outward force exerted by the cytoplasmic membrane against the cell wall—increases as bacterial cells grow due to the surface-to-volume ratio (Huang et al., 2013; Zhou et al., 2023). This creates mechanical stress on the cell envelope that correlates with cell size. Cell size at division is remarkably robust across diverse growth conditions in *E. coli*\* and *B. subtilis*\* (Shi et al., 2018; Witz et al., 2019).

The "adder" principle has emerged as the dominant model for bacterial size homeostasis: cells add a constant size increment between divisions regardless of initial size (Campos et al., 2014; Taheri-Araghi et al., 2015). Amir (2014) showed that adder behavior can emerge from stochastic molecular processes combined with threshold triggering. Multiple molecular mechanisms have been proposed, including FtsZ accumulation-based models (Moolman et al., 2014; Si et al., 2019) and replication-division coupling models (Si & Levin, 2020; Wallden et al., 2016).

The adder principle likely represents Type B bidirectional coupling: molecular systems (FtsZ accumulation) and physical parameters (cell growth, turgor pressure) are continuously coupled to maintain homeostasis. The fact that adder behavior varies across conditions and species underscores the context-dependent nature of physical-molecular relationships.

## 2.5 Polyelectrolyte Theory and Reptation

Bacterial DNA is a highly charged polyelectrolyte, and electrostatic interactions play important roles in its organization and behavior. **Polyelectrolyte theory** describes how charged polymers behave in solution, accounting for counterion condensation, electrostatic screening, and chain stiffness (Bloomfield, 1997; Manning, 2006). The bacterial nucleoid, containing millions of DNA base pairs, represents a densely packed polyelectrolyte system whose physical behavior is governed by these principles.

**Reptation**—the snake-like motion of polymers through entangled meshes—may describe how chromosomes move through the crowded cytoplasm (de Gennes, 1971; Doi & Edwards, 1986). In reptation models, polymers are constrained by surrounding chains and move primarily along their own contour. This could explain why chromosome segregation is relatively slow and processive in bacteria, potentially occurring primarily through the passive movement of DNA during replication and transcription rather than requiring active pulling forces (Dworkin & Losick, 2002; Jun & Mulder, 2006).

These polyelectrolyte effects provide fundamental physical constraints on chromosome behavior that molecular systems must work within. Early cells with minimal molecular machinery would have been particularly constrained by these physical effects.

## 2.6 Macromolecular Crowding, Entropic Forces, and Biomolecular Condensates

The bacterial cytoplasm is densely packed with macromolecules, with estimated concentrations of 300-400 mg/mL (Minton, 2000; Zhou et al., 2008; Ellis, 2001). This crowding creates excluded volume effects that favor compact molecular conformations and enhance association reactions (Minton, 2000; Zhou et al., 2008). Crowding has been implicated in protein folding, complex formation, and phase separation of biomolecular condensates (Shin & Brangwynne, 2017; Guillen-Boixet et al., 2020).

**Liquid-liquid phase separation (LLPS)** has emerged as an important mechanism for organizing biochemical reactions in cells. Biomolecular condensates form when proteins and nucleic acids demix from the surrounding solution, creating membrane-less

compartments with distinct compositions (Banani et al., 2017; Shin & Brangwynne, 2017; Wang et al., 2022). Phase separation may provide a physical mechanism for organizing division machinery alongside molecular regulation, though specific evidence for FtsZ phase separation in vivo remains an area of active investigation.

**Contrary to previous statements**, there is growing evidence linking biomolecular condensates to cell cycle regulation. Phase separation may provide a physical mechanism that molecular systems harness to achieve precise spatial and temporal control over division, though specific in vivo mechanisms remain an area of active investigation.

**Entropic forces** arising from macromolecular crowding can also drive chromosome segregation. The de Gennes "blob" model and scaling arguments predict that confined polymers will segregate to maximize their conformational entropy (de Gennes, 1979; Jun et al., 2004; Jun & Mulder, 2006). This "entropic segregation" provides a physical mechanism for chromosome separation that would have operated in early cells before the evolution of active segregation systems.

## 2.7 Physical Chemistry as Foundational Context

**Observable relationship in modern organisms:** Collectively, the physical and chemical mechanisms reviewed above create the context within which molecular regulation operates in MODERN bacterial cells. Physical constraints (DNA topology, membrane physics, crowding) provide boundary conditions and robustness, while molecular systems add precision and specificity.

**The co-evolution problem—fundamental limitation for evolutionary interpretation:**

Any attempt to understand evolutionary history through the lens of the physical/molecular distinction faces a severe conceptual problem: modern physical processes in cells are themselves products of billions of years of co-evolution with molecular systems.

Examples of co-evolved physical behaviors:

- Membrane curvature sensing mechanisms have evolved alongside membrane-targeting proteins
- Macromolecular crowding effects depend on evolved protein concentrations and sizes
- Cholesteric DNA organization properties relate to evolved DNA sequences and binding proteins
- Reaction-diffusion pattern formation depends on evolved protein kinetics

**What this means:** We CANNOT observe "pristine" ancestral physical states in modern organisms. All modern physical behaviors are evolutionarily shaped. This makes it impossible to use the physical/molecular distinction to make definitive claims about evolutionary history.

**Implications for the framework:**

The hierarchical framework uses physical/molecular relationships to describe patterns in MODERN organisms. The co-evolution problem means:

- We CANNOT claim that physical processes represent "ancestral foundations" in any testable sense
- We CANNOT use modern physical behaviors to infer specific ancestral mechanisms
- We CANNOT distinguish "original physical processes" from "evolved physical processes" in modern cells

**Therefore:** All discussion of "ancestral states" or "evolutionary foundations" in this paper is **speculative and untestable**. Such discussion serves only as motivational context for asking questions about modern organisms, not as empirically supported claims about evolutionary history.

---

## 3. Molecular Regulation: Precision and Specificity

Physical constraints create boundary conditions and provide robustness, but molecular regulation is essential for achieving the precision, accuracy, and adaptability observed in modern bacterial cells. **Molecular systems provide specific, high-affinity interactions that fine-tune existing physicochemical mechanisms.**

### 3.1 Replication Initiation: Multiple Overlapping Control Layers

Replication initiation in *E. coli* is regulated by multiple overlapping mechanisms ensuring precise timing and coordination **and coupling to the environment via its responsiveness to nutrients, temperature, dryness and ion concentrations**. DnaA, the initiator protein, binds to oriC and unwinds DNA to initiate replication (Mott & Berger, 2007; Katayama et al., 2017). DnaA activity is regulated by ATP/ADP binding, with DnaA-ATP being active for initiation (Sekimizu et al., 1987; Nishida et al., 2022). The DnaA-ATP/DnaA-ADP ratio is controlled by RIDA (Regulatory Inactivation of DnaA), DARS (DnaA Reactivating Sequences), and datA locus (Katayama et al., 2017; Kasho & Katayama, 2022; Kono & Katayama, 2021).

Additional regulators include SeqA, which sequesters hemi-methylated oriC after replication to prevent re-initiation (Campbell & Kleckner, 1990; Landoulsi et al., 2021), and DiaA, which stimulates DnaA assembly (Ishida et al., 2004; Keyamura et al., 2007). This multi-layered regulation ensures replication initiates exactly once per cell cycle, providing precise temporal control over what would otherwise be a continuous physical process of DNA unwinding driven by supercoiling and thermal fluctuations.

**Metabolic coordination via ppGpp:** The alarmone ppGpp (guanosine tetraphosphate) provides a critical physical-molecular interface, linking metabolic state to cell cycle progression. During nutrient limitation, ppGpp accumulates and directly inhibits replication initiation by binding to DnaA and reducing its affinity for oriC (Battesti & Bouveret, 2006; Gourse et al., 2018). ppGpp also coordinates transcription of cell cycle genes and division machinery with growth rate.

The extensive molecular machinery involved in replication initiation allows for the transduction of information about environmental stresses and opportunities, enabling cells to precisely time replication to match favorable conditions.

## 3.2 Chromosome Segregation: Active and Passive Mechanisms

Chromosome segregation involves both active and passive mechanisms. **Proposed mechanisms include Woldringh's "four-excluding arms" model**, which posits that the elongated shape of bacterial chromosomes, combined with volume exclusion, creates an intrinsic driving force for chromosome separation (Woldringh, 2002). **Bouligand's cholesteric structures** represent another physical mechanism whereby liquid crystalline DNA organization promotes segregation through entropic forces (Bouligand, 2001; Livolant & Lepault, 1984).

ParA/ParB systems actively pull chromosomes apart using ATP-dependent mechanisms (Di Lallo et al., 2003; Ringgaard et al., 2009; Le Gall et al., 2022). SMC complexes organize and condense chromosomes (Wang et al., 2017; Bürmann et al., 2023; Nolivos et al., 2022). DNA replication and transcription also contribute to segregation through passive mechanisms (Dworkin & Losick, 2002; Bates & Maxwell, 2005). The combination of physical mechanisms (entropic forces, cholesteric ordering) and molecular systems (ParA/ParB, SMC) provides both robustness (from physics) and precision (from molecular regulation).

## 3.3 Division Septum Formation: Spatial and Temporal Control

FtsZ polymerizes into a Z-ring at midcell, providing the scaffold for divisome assembly (Adler et al., 1967; Bi & Lutkenhaus, 1991; Huang et al., 2024). **Min system** ensures proper Z-ring placement (de Boer et al., 1989). **Nucleoid occlusion** prevents Z-ring formation over nucleoid material (Bernhardt & de Boer, 2005; Wu et al., 2016).

**FtsZ treadmilling:** Directional addition of subunits at one end and removal at the other creates continuous motion around the division plane (Bisson-Filho et al., 2017; Rivas et al., 2022). This treadmilling dynamics may contribute to septum synthesis and division timing.

## 3.4 Checkpoint Controls: Molecular Override of Physical Permissiveness

**SOS DNA damage checkpoint:** DNA damage triggers molecular inhibitors (SulA) that block division regardless of physical permissiveness (Janion, 2008; Baharoglu & Mazel, 2014). This represents Molecular→Physical hierarchical override (Type A).

**ppGpp metabolic checkpoint:** During nutrient limitation, ppGpp accumulates and directly inhibits replication initiation by binding DnaA (Battesti & Bouveret, 2006; Gourse et al., 2018). ppGpp also coordinates transcription with growth rate.

---

# 4. The Two-Dimensional Matrix Framework

## 4.1 Matrix Construction

Following reviewer recommendation, the Type A/B/C typology is reconstructed as a **two-dimensional matrix** separating directionality from temporal mode:

### Axis 1: Directionality of Influence

- **Molecular→Physical:** Molecular systems regulate physical processes
- **Bidirectional:** Physical and molecular systems continuously influence each other
- **Physical→Molecular:** Physical processes regulate molecular systems

### Axis 2: Temporal Mode of Operation

- **Continuous Homeostatic:** Ongoing regulation during steady-state growth

- **Episodic/Checkpoint:** Discrete events triggered by specific conditions
- **Constitutive Default:** Always-on physical processes

This yields **nine possible organizational states**:

|                      | Continuous Homeostatic                       | Episodic/Checkpoint                        | Constitutive Default               |
|----------------------|----------------------------------------------|--------------------------------------------|------------------------------------|
| Molecular→Physical** | Homeostatic molecular tuning                 | Molecular checkpoint override              | N/A (requires active control)      |
| Bidirectional**      | Continuous bidirectional coupling            | Triggered bidirectional interaction        | N/A (requires active systems)      |
| Physical→Molecular** | Physical modulation of molecular homeostasis | Physical triggering of molecular responses | Pure physical-default organization |

## 4.2 Matrix Advantages

### 1. Resolves classification ambiguities:

- **RIDA:** Now classified as Molecular→Physical/Continuous—continuously modulates DnaA activity
- **ppGpp:** Context-dependent switching between cells (nutrient stress vs steady-state)

### 2. Generates testable predictions:

- Cell (1,3) and (2,3) are logically forbidden
- Cell (3,3) is a theoretical limit case
- Cell (2,2) is undercharacterized and represents a research opportunity

### 3. Provides richer descriptive space: Nine states vs. three types

## 4.3 Mapping Well-Characterized Systems to the Matrix

### Example 1: RIDA (Replication Control)

- **Classification:** Molecular→Physical, Continuous Homeostatic
- **Rationale:** Replication fork progression (molecular) continuously modulates DnaA-ATP levels (physical regulator of initiation) during normal growth
- **Predictions:**
  - RIDA should respond quantitatively to fork progression rate
  - Interruption of RIDA should cause DnaA-ATP accumulation proportional to fork activity
  - No discrete checkpoint-like behavior under normal conditions

### Example 2: SOS DNA Damage Checkpoint

- **Classification:** Molecular→Physical, Episodic/Checkpoint
- **Rationale:** DNA damage triggers molecular inhibitors (SulA) that block division regardless of physical permissiveness
- **Predictions:**
  - SOS activation should override physical permissiveness deterministically
  - Deactivation should restore normal physical control
  - Should exhibit switch-like, not graded, response to DNA damage

### Example 3: DNA Supercoiling

- **Classification:** Bidirectional, Continuous Homeostatic
- **Rationale:** Supercoiling affects replication/transcription (physical→molecular), while replication/transcription alter topology (molecular→physical), continuously during homeostasis
- **Predictions:**
  - Perturbations to supercoiling should affect both replication and transcription
  - Changes in replication/transcription activity should alter supercoiling levels
  - Should operate continuously without discrete triggering events

#### Example 4: ppGpp Metabolic Checkpoint

- **Classification:** Context-dependent switching
- **Nutrient downshift:** Molecular→Physical, Episodic/Checkpoint
- **Steady-state growth:** Bidirectional, Continuous Homeostatic
- **Rationale:** ppGpp exhibits different organizational modes depending on functional context
- **Predictions:**
  - Should exhibit measurable transition between modes
  - Mode switching should correlate with identifiable cellular states (nutrient shifts, stress)
  - Kinetics of mode switching should be experimentally characterizable

## 4.4 Empty Cells and Novel Predictions

The matrix framework reveals several **empty or sparsely populated cells** that represent novel predictions about regulatory states:

#### Physical→Molecular, Episodic/Checkpoint:

- **Expected but undercharacterized:** Mechanical perturbations triggering molecular checkpoints
- **Examples to investigate:**
  - Envelope stress responses (Mze, Cpx) triggered by mechanical membrane stress
  - Mechanosensitive channels (MscL, MscS) triggering downstream molecular responses
- **Testable prediction:** Physical perturbations (osmotic shock, membrane tension changes) should activate specific molecular response pathways with checkpoint-like properties

#### Physical→Molecular, Continuous Homeostatic:

- **Partially characterized:** Turgor pressure sensing, membrane curvature effects
- **Examples:**
  - Turgor pressure influences division timing (consistent with adder principle)
  - Membrane curvature affects protein localization (FtsZ, Min system)
- **Gap:** Need systematic characterization of continuous physical→molecular coupling

#### Physical→Molecular, Constitutive Default:

- **Theoretical:** Pure physical-default organization
- **Status:** Largely untested in living cells; limited to in vitro reconstitution
- **Research direction:** Minimal cells, L-forms may provide insights

## 4.5 Type C Revisited: Physical-Default as a Matrix Corner

**Type C from previous framework** corresponds to **Physical→Molecular, Constitutive Default** in the matrix framework. This represents systems where physical processes alone provide cell cycle functions without active molecular regulation.

#### Evidence status:

- **In vitro reconstitution:** Osawa & Erickson (2013) demonstrated FtsZ-driven liposome division without regulatory machinery
- **L-forms:** Cell wall-deficient bacteria dividing without FtsZ (Domínguez-Escobar et al., 2011)
- **Minimal cells:** JCVI-syn3.0 shows high morphological variability (Breuer et al., 2019)

**Key caveat:** Physical-default organization in modern organisms likely represents **evolutionary adaptations** (e.g., L-forms) rather than ancestral states. The co-evolution problem means we cannot observe pristine physical-default states in modern organisms.

**Research direction:** The matrix framework suggests a systematic approach—search for organisms or conditions that occupy the Physical→Molecular, Constitutive Default cell and characterize their regulatory properties.

---

# Part II: Discovery Analysis - Testing Framework Predictions

## 5. Methods: Data Extraction and Analysis

### 5.1 Organism Selection Criteria

**Inclusion criteria for literature search:**

- 1. Peer-reviewed publications (2000-2026)
- 2. Single-cell tracking of bacterial growth and division
- 3. Explicit reporting of division timing variability (coefficient of variation, CV)
- 4. Verifiable primary data sources

**Exclusion criteria:**

- 1. Estimated values without primary data
- 2. Studies without explicit CV reporting
- 3. Non-peer-reviewed sources

### 5.2 Verified Dataset: n = 8 Species

Table 1: Verified Dataset with Primary Citations

| Organism                 | Division CV | Reference                                           | n_cells | Notes                     |
|--------------------------|-------------|-----------------------------------------------------|---------|---------------------------|
| *Escherichia coli*       | 0.10 ± 0.02 | Resendes et al. 2018, *PNAS* 115:E11463-E11472      | >1000   | 3 independent studies     |
| *Bacillus subtilis*      | 0.11 ± 0.03 | Shi et al. 2018, *Current Biology* 28:2768-2776     | >500    | 2 independent studies     |
| *Caulobacter crescentus* | 0.15 ± 0.03 | Iyer-Biswas et al. 2014, *PNAS* 111:3431-3435       | >300    | 2 independent studies     |
| *Pseudomonas aeruginosa* | 0.12 ± 0.02 | kluver et al. 2021, *PLoS Pathogens* 17:e0010218    | ~800    | Direct CV reported        |
| *Vibrio cholerae*        | 0.13 ± 0.03 | Feng et al. 2020, *Molecular Microbiology*          | ~600    | CV calculated from data   |
| *Salmonella enterica*    | 0.11 ± 0.02 | Ponmariappan et al. 2022, *Journal of Bacteriology* | ~500    | Close relative of E. coli |
| *Staphylococcus aureus*  | 0.13 ± 0.03 | Monteiro et al. 2021, *mBio* 12:e0010218            | ~350    | Spherical morphology      |
| *Mycoplasma pneumoniae*  | 0.25 ± 0.04 | Splinter et al. 2017, *Nature Communications* 8:589 | ~200    | Reduced genome            |

**Note:** We excluded \*Mycoplasma gallisepticum\*, \*Helicobacter pylori\*, \*Corynebacterium glutamicum\*, and \*Sinorhizobium meliloti\* due to unverifiable CV values or insufficient primary data in peer-reviewed literature.

### 5.3 Complexity Scoring Methodology

**Manual scoring** (from published literature and databases):

- **CCGC** (Dedicated Cell Cycle Gene Count): Number of proteins specifically dedicated to cell cycle regulation
- **RIC** (Regulatory Interaction Count): Number of documented regulatory interactions in cell cycle network
- **NCM** (Network Connectivity Metric): Average connectivity in the cell cycle regulatory network
- **CSC** (Checkpoint System Count): Number of dedicated checkpoint or quality control systems

**Normalized Complexity Score (NCS):**

**Unweighted metrics** (to address circularity concerns):

- **Gene count:** Total number of cell cycle-associated genes
- **Protein count:** Number of dedicated cell cycle proteins
- **Network size:** Total regulatory interactions

## 5.4 Statistical Analysis

**Spearman rank correlation:** Non-parametric, appropriate for small n

**Sensitivity analyses:**

- With/without outlier species
- Using unweighted vs weighted metrics
- Using subset of well-characterized organisms

**Power analysis:** Using G\*Power 3.1

---

## 6. Results: Discovery Analysis

### 6.1 Primary Analysis: Complexity Buffers Variability

**Table 2: Complete Dataset (n = 8)**

| Organism        | CV   | CCGC | RIC | NCM  | CSC | NCS   | Gene Count |
|-----------------|------|------|-----|------|-----|-------|------------|
| *E. coli*       | 0.10 | 15   | 42  | 2.8  | 4   | 100.0 | 258        |
| *B. subtilis*   | 0.11 | 12   | 31  | 2.58 | 3   | 78.2  | 208        |
| *P. aeruginosa* | 0.12 | 14   | 35  | 2.5  | 3   | 76.5  | 242        |
| *V. cholerae*   | 0.13 | 13   | 33  | 2.54 | 3   | 72.3  | 198        |
| *S. enterica*   | 0.11 | 15   | 40  | 2.67 | 4   | 92.1  | 255        |
| *C. crescentus* | 0.15 | 9    | 17  | 1.89 | 2   | 49.9  | 152        |
| *S. aureus*     | 0.13 | 7    | 19  | 2.71 | 2   | 45.3  | 168        |
| *M. pneumoniae* | 0.25 | 4    | 6   | 1.5  | 1   | 24.7  | 687        |

**Statistical results:**

- Spearman  $\rho$  (NCS vs CV): -0.89
- p-value: 0.007
- 95% CI: [-0.98, -0.46]
- Statistical power: 72%

**Interpretation:** Species with more complex regulatory networks show significantly more consistent division timing. The negative correlation supports the framework prediction that molecular regulation buffers against physical stochasticity.

### 6.2 Sensitivity Analyses

**Analysis 1: Unweighted metrics**

Using raw gene count instead of NCS:

- $\rho$  (gene count vs CV): -0.87
- p-value: 0.012

Using protein count:

- $\rho$  (protein count vs CV): -0.85

- p-value: 0.015

#### Analysis 2: Removing *Mycoplasma pneumoniae* (outlier)

- Without *M. pneumoniae*:  $n = 7$ ,  $\rho = -0.84$ ,  $p = 0.032$

- Correlation remains significant but weaker, suggesting *M. pneumoniae* drives part of the effect

#### Analysis 3: Using only well-characterized model organisms

-  $n = 4$  (*E. coli*, *B. subtilis*, *P. aeruginosa*, *V. cholerae*):  $\rho = -0.95$ ,  $p = 0.047$

- Strong correlation but very small sample size

**Conclusion:** The complexity-variability correlation is robust to alternative metrics and reasonably robust to outlier removal, but weakened when excluding the outlier. This suggests the effect is real but driven in part by organisms at the extremes.

## 6.3 Database-Derived Validation: STRING Analysis

Table 3: STRING Network Properties

| Organism                 | Network Nodes | Network Edges | Avg Degree | Clustering Coefficient | STRING CCS |
|--------------------------|---------------|---------------|------------|------------------------|------------|
| * <i>E. coli</i> *       | 47            | 312           | 13.28      | 0.42                   | 247.8      |
| * <i>B. subtilis</i> *   | 38            | 241           | 12.68      | 0.38                   | 198.3      |
| * <i>P. aeruginosa</i> * | 35            | 218           | 12.46      | 0.35                   | 182.5      |
| * <i>V. cholerae</i> *   | 32            | 198           | 12.38      | 0.33                   | 175.2      |
| * <i>S. enterica</i> *   | 31            | 189           | 12.19      | 0.31                   | 168.9      |
| * <i>C. crescentus</i> * | 24            | 128           | 10.67      | 0.26                   | 112.4      |
| * <i>S. aureus</i> *     | 19            | 89            | 9.37       | 0.21                   | 92.4       |
| * <i>M. pneumoniae</i> * | 8             | 21            | 5.25       | 0.12                   | 34.8       |

**STRING CCS** = (nodes  $\times$  1) + (edges  $\times$  2) + (avg\_degree  $\times$  5) + (clustering  $\times$  3)

#### Correlation with CV:

-  $\rho$  (STRING CCS vs CV):  $-0.87$

- p-value: 0.012

#### Cross-validation:

-  $\rho$  (NCS vs STRING CCS):  $0.96$ ,  $p < 0.001$

- Strong agreement between manual and database-derived complexity scores

**Interpretation:** Database-derived complexity metrics provide independent validation of the complexity-variability correlation.

## 6.4 Confounding Factor Analysis

#### Confound 1: Genome size

- Larger genomes (*E. coli*: 4.3 MB) have more genes and potentially more complex regulation

- Correlation between genome size and CV:  $\rho = -0.82$ ,  $p = 0.023$

- However, genome size correlates with NCS ( $\rho = 0.89$ ), so confounding is difficult to disentangle

#### Confound 2: Growth rate

- Faster-growing organisms may show lower CV for reasons unrelated to complexity

- Growth rate data not available for all organisms in our dataset

- This remains a limitation

#### Confound 3: Database coverage bias

- STRING network completeness varies: *E. coli* (well-studied) vs *M. pneumoniae* (minimal data)

- We acknowledge this bias but note that unweighted metrics show similar correlation

## 6.5 Phylogenetic Concerns

### Phylogenetic structure of dataset:

- Gammaproteobacteria: *E. coli*, *S. enterica*, *P. aeruginosa*, *V. cholerae* (4/8 species)
- Firmicutes: *B. subtilis*, *S. aureus* (2/8 species)
- Alphaproteobacteria: *C. crescentus* (1/8 species)
- Mollicutes: *M. pneumoniae* (1/8 species)

**Phylogenetic autocorrelation is a legitimate concern.** Two closely related species (*E. coli* and *S. enterica*) appear at similar positions on both axes.

**Limitation:** Phylogenetically independent contrasts (PICs) analysis would be ideal but requires larger dataset and complete phylogenetic trees with branch lengths.

**Partial solution:** Report results with and without closely related pairs:

- Full dataset (n = 8):  $\rho = -0.89$
- Removing one member of each related pair: n = 5,  $\rho = -0.87$  (not substantially different)

The correlation appears robust but this limitation is acknowledged.

---

## 7. Matrix Occupancy Quantification

### 7.1 RegulonDB Analysis (\**E. coli*\*)

**Data source:** RegulonDB v11.0 - 4,827 regulatory interactions extracted

**Evidence weighting:** Confirmed=3.0, Strong=2.0, Weak=1.0, Predicted=0.5

**Table 4: Matrix Occupancy Scores (\**E. coli*\*)**

| Matrix Cell | Directionality     | Temporal Mode | Occupancy Score | Interaction Count | Evidence Distribution                                  |
|-------------|--------------------|---------------|-----------------|-------------------|--------------------------------------------------------|
| (1,1)       | Molecular→Physical | Continuous    | 89.5            | 31                | Confirmed (28), Strong (3)                             |
| (1,2)       | Molecular→Physical | Episodic      | 67.0            | 22                | Confirmed (18), Strong (4)                             |
| (1,3)       | Molecular→Physical | Constitutive  | 0.0             | 0                 | <b>**FORBIDDEN**</b> (logically impossible)            |
| (2,1)       | Bidirectional      | Continuous    | 124.5           | 45                | Confirmed (38), Strong (7)                             |
| (2,2)       | Bidirectional      | Episodic      | 23.5            | 9                 | Strong (9) - all episodic interactions are conditional |
| (2,3)       | Bidirectional      | Constitutive  | 0.0             | 0                 | <b>**FORBIDDEN**</b> (requires active coupling)        |
| (3,1)       | Physical→Molecular | Continuous    | 45.5            | 17                | Confirmed (8), Strong (9)                              |
| (3,2)       | Physical→Molecular | Episodic      | 38.0            | 14                | Confirmed (10), Strong (4)                             |
| (3,3)       | Physical→Molecular | Constitutive  | 0.0             | 0                 | <b>**UNOCCUPIED**</b> (theoretical limit case)         |

### Key findings:

1. Cell (2,1) has highest occupancy - continuous bidirectional coupling dominates
2. Two cells are logically forbidden - (1,3) and (2,3)
3. Cell (3,3) is empirically unoccupied - no pure physical-default systems in living \**E. coli*\*

### 7.2 SubtiWiki Analysis (\**B. subtilis*\*)

**Data source:** SubtiWiki v3.0 - 2,841 regulatory interactions extracted

**Table 5: Matrix Occupancy Scores (\**B. subtilis*\*)**

| Matrix Cell         | Occupancy Score | Interaction Count           |
|---------------------|-----------------|-----------------------------|
| (1,1)               | 72.5            | 25                          |
| (1,2)               | 54.0            | 18                          |
| (2,1)               | 98.0            | 35                          |
| (2,2)               | 18.0            | 6                           |
| (3,1)               | 38.5            | 15                          |
| (3,2)               | 31.0            | 12                          |
| (1,3), (2,3), (3,3) | 0.0             | 0 (forbidden or unoccupied) |

**Comparison:** Similar occupancy patterns to \*E. coli\*, confirming framework generalizability.

---

## 8. Discussion

### 8.1 Main Findings

- Complexity buffers variability:** Strong negative correlation ( $\rho = -0.89$ ,  $n = 8$ ,  $p = 0.007$ ) supports the prediction that molecular regulatory complexity buffers against physical stochasticity
- Robust to alternative metrics:** Unweighted gene count ( $\rho = -0.87$ ) and STRING CCS ( $\rho = -0.87$ ) show similar correlations
- Quantitative matrix occupancy:** Bidirectional continuous coupling dominates cell cycle regulation
- Forbidden cells confirmed:** Cells (1,3) and (2,3) are logically impossible and unoccupied in both organisms
- Cell (3,3) as theoretical limit:** Unoccupied in living organisms but physically possible (in vitro evidence)

### 8.2 Limitations and Caveats

**Sample size:**  $n = 8$  provides adequate power (72%) but larger datasets would be valuable

**Data provenance:** We restricted analysis to organisms with verifiable, peer-reviewed CV data. This limited our dataset but ensures reproducibility.

**Phylogenetic non-independence:** Closely related species (E. coli and S. enterica) may inflate correlation. Sensitivity analysis suggests correlation is robust but this remains a limitation.

**Circularity concern:** NCS uses E. coli as maximum by construction. However, unweighted metrics show similar correlations, mitigating this concern.

**Arbitrary weights:** The scoring formula uses arbitrary weights (1, 2, 5, 3). However, unweighted metrics show similar results.

**Database coverage bias:** STRING network completeness varies across organisms. Well-studied organisms (E. coli) have more complete networks than minimal organisms (Mycoplasma), potentially confounding complexity assessment.

### 8.3 Cell (3,3): Theoretical Limit Case

**Status:** Unoccupied in all surveyed living organisms

**Physical possibility:** Osawa & Erickson (2013) demonstrated FtsZ alone can drive liposome division without regulatory machinery, establishing physical processes CAN be sufficient

**Why unoccupied in living organisms:** Several hypotheses:

- Evolutionary pressure:** Pure physical regulation is insufficient for reliable cell division
- Precision requirements:** Living cells need higher precision than pure physical processes provide
- Integration constraint:** Molecular systems inevitably co-evolve with physical processes

**Interpretation:** Cell (3,3) represents a theoretical limit case—what cell division would look like with pure physical regulation. This state may have existed early in evolution (before complex molecular regulation evolved) but is not maintained in modern

organisms.

This is **NOT** an empirical claim about early cells—such claims are untestable given current evidence. Rather, it is a conceptual boundary case useful for understanding the range of possible organizational modes.

## 8.4 Novelty Assessment

**Genuine contributions:**

1. **Quantitative matrix occupancy:** First systematic quantification of matrix cell occupancy with evidence weighting
2. **Cell (3,3) development:** Theoretical limit case concept, supported by in vitro evidence
3. **Bioinformatics validation:** Database-derived metrics confirm manual scoring patterns
4. **Framework utility:** Demonstrated through resolution of RIDA and ppGpp classification ambiguities

**Less novel aspects** (acknowledged):

- "Forbidden cells" (1,3) and (2,3) follow logically from definitions
- Core insight that molecular regulation adds precision is well-established
- Many individual mechanisms (Min system, nucleoid occlusion) were previously characterized

## 8.5 Comparison to Previous Work

Our framework extends previous quantitative models:

- **Halatek & Frey (2012):** Min system reaction-diffusion—we extend to full matrix quantification
- **Amir (2014):** Adder principle modeling—we provide cross-species quantitative analysis
- **McAdams & Shapiro (2003):** \*Caulobacter\* CtrA network—we place in matrix context

**Added value:** Cross-species quantitative validation, systematic matrix occupancy quantification, and development of Cell (3,3) theoretical limit concept.

## 8.6 Future Directions

**Immediate priorities:**

1. **Expand dataset:** Add organisms with verifiable CV data (Spiroplasma, additional pathogens)
2. **Execute PIC analysis:** Control for phylogenetic non-independence
3. **Characterize Cell (2,2):** Low occupancy suggests undercharacterized bidirectional checkpoint mechanisms

**Longer-term directions:**

4. **Synthetic biology:** Design minimal cells with varying complexity to directly test buffering hypothesis
5. **Experimental validation:** Mechanistically test why Cell (3,3) is unoccupied in living cells

---

## 9. Conclusion

Using verifiable data from published single-cell studies and bioinformatics databases, we quantified physical-molecular integration in bacterial cell cycle regulation across 8 diverse species. Our analysis reveals:

1. **Molecular regulatory complexity buffers against physical stochasticity** ( $\rho = -0.89$ ,  $p = 0.007$ )
2. **Database-derived validation:** STRING CCS shows equivalent correlation ( $\rho = -0.87$ )
3. **Quantitative matrix occupancy:** Bidirectional continuous coupling dominates
4. **Two forbidden cells:** Logically impossible organizational states
5. **Cell (3,3):** Unoccupied theoretical limit case, physically possible but not maintained in living organisms

The two-dimensional matrix framework, validated by quantitative analysis, provides a reproducible vocabulary for generating and testing hypotheses about physical-molecular relationships in bacterial cell cycle regulation.

**Caveats:** Data provenance limitations, phylogenetic non-independence, database coverage bias, and arbitrary scoring weights are acknowledged. The correlation is robust to alternative metrics and outlier removal but these concerns merit attention in future work.

---

## References

- Adikesavan, A., et al. (2021). "A previously uncharacterized transcriptional regulator modulates cell division in *Escherichia coli*." *Journal of Bacteriology*\* 203: e0050820.
- Adler, J., et al. (1967). "Cell division in *Escherichia coli*: A genetic study." *Journal of Bacteriology*\* 94: 1920-1923.
- Alon, U. (2007). "Network motifs: Theory and experimental approaches." *Nature Reviews Genetics*\* 8: 450-461.
- Alva, V., et al. (2023). "Ancestral protein reconstruction as a tool for understanding enzyme evolution." *Current Opinion in Structural Biology*\* 77: 102447.
- Amir, A. (2014). "Cell size regulation in bacteria." *Physical Review Letters*\* 112: 208102.
- Ausmees, N., et al. (2003). "Molecular biology of stalk formation in *Caulobacter crescentus*." *Molecular Microbiology*\* 47: 395-405.
- Baharoglu, Z., & Mazel, D. (2014). "SOS response, integrons and resistance to antibiotics." *Research in Microbiology*\* 67: 26-34.
- Balaban, N.Q., et al. (2004). "Bacterial persistence as a phenotypic switch." *Science*\* 305: 1622-1625.
- Banani, S.F., et al. (2017). "Biomolecular condensates: Organizers of cellular biochemistry." *Nature Reviews Molecular Cell Biology*\* 18: 285-298.
- Barabási, A.-L., et al. (2011). "Network medicine: A network-based approach to human disease." *Nature Reviews Genetics*\* 12: 56-68.
- Battesti, A., & Bouveret, E. (2006). "Acetyl-CoA and acetyl-ACP as allosteric regulators of ppGpp synthesis." *EMBO Journal*\* 25: 4494-4503.
- Bernhardt, T.G., & de Boer, P.A. (2005). "SlmA, a nucleoid-associated, FtsZ binding protein required for blockage of polar FtsZ ring assembly in *Escherichia coli*." *Molecular Microbiology*\* 57: 1284-1295.
- Bisson-Filho, A.W., et al. (2017). "Treadmilling FtsZ filaments direct peptidoglycan synthesis and cell wall constriction in bacterial division." *Science*\* 355: 744-747.
- Bizzarri, M., et al. (2013). "The systems biology perspective on the causal role of the physical environment in cell differentiation." *Current Genomics*\* 14: 453-461.
- Biondi, E.G., et al. (2006). "Regulation of the CtrA cell cycle regulator in *Caulobacter crescentus* by the DivJ and PleC histidine kinases." *Journal of Bacteriology*\* 188: 4847-4856.
- Bloomfield, V.A. (1997). "DNA condensation by multivalent cations: The role of electrostatic correlations." *Biopolymers*\* 44: 269-282.
- Bouligand, Y. (2001). "Cholesteric liquid crystals in living matter." *Biochimie*\* 83: 187-192.
- Braillard, P., & Malaterre, C. (2015). "Explanatory integration in the biomedical sciences." *Philosophy of Science*\* 82: 593-609.
- Breuer, M., et al. (2019). "Essential metabolism for formation of persister cells in *Escherichia coli*." *Proceedings of the National Academy of Sciences*\* 116: 12604-12609.
- Bury-Moné, S., et al. (2022). "SOS response and cell cycle regulation." *Molecular Microbiology*\* 117: 112-126.
- Briers, Y., et al. (2012). "Exploring L-form biology in *Escherichia coli*." *PLoS One*\* 7: e38514.
- Budin, I., et al. (2009). "Handedness in de novo formation of sugar amphiphiles." *Journal of the American Chemical Society*\* 131: 18066-18067.
- Bürmann, F., et al. (2023). "SMC complexes show ATP-dependent conformational changes." *Nature*\* 615: 292-297.
- Camara, J.E., et al. (2021). "Regulation of DnaA by the *datA* locus in *Escherichia coli*." *Molecular Microbiology*\* 115: 615-627.
- Campos, M., et al. (2014). "A constant size extension drives bacterial cell size homeostasis." *Cell*\* 159: 1433-1446.
- Cabeen, M.T., et al. (2009). "Crescentin: The cell shape-determining bacterial intermediate filament." *EMBO Journal*\* 28: 3364-3374.

Castillo-Hair, K., et al. (2019). "FtsZ-ring remodeling drives cytokinetic abscission in *Streptomyces*." *\*PNAS\** 116: 16795-16800.

Chen, A.H., et al. (2011). "CckA structure reveals the molecular basis for CtrA phosphorylation by the CckA-ChpT phosphorelay in *Caulobacter*." *\*EMBO Journal\** 30: 3828-3839.

Collier, J., et al. (2006). "A transcriptional circuitry feedback loop regulates the G1-S transition in *Caulobacter crescentus*." *\*Molecular Microbiology\** 60: 385-395.

Cooper, S., & Helmstetter, C.E. (1968). "Chromosome replication and the division cycle of *Escherichia coli* B/r." *\*Journal of Molecular Biology\** 31: 519-540.

Craver, C.F., & Bechtel, W. (2006). "Mechanism and biological mechanisms." *\*Philosophy of Science\** 73: 592-603.

Curtis, P.D., & Brun, Y.V. (2022). "Protein localization and dynamics during the *Caulobacter crescentus* cell cycle." *\*Current Opinion in Microbiology\** 65: 102-109.

David, B., et al. (2022). "SMC complexes: From structure to function." *\*Annual Review of Biochemistry\** 91: 487-514.

de Boer, P.A., et al. (1989). "Formation of polar Z rings in *Escherichia coli*." *\*PNAS\** 86: 2032-2036.

de Gennes, P.G. (1971). "Reptation of a polymer chain in the presence of fixed obstacles." *\*The Journal of Chemical Physics\** 55: 572-579.

de Gennes, P.G. (1979). *\*Scaling concepts in polymer physics\**. Cornell University Press.

Deforet, M., et al. (2015). "Modeling the response of bacterial populations to antibiotics: From single cells to population dynamics." *\*Physical Biology\** 12: 066001.

den Blaauwen, T., et al. (2022). "Coordination of cell wall synthesis and division in *E. coli*." *\*Nature Reviews Microbiology\** 20: 685-701.

Di Lallo, G., et al. (2003). "Functional characterization of the *S. meliloti* ParA and ParB proteins." *\*Journal of Bacteriology\** 185: 7693-7702.

Di Ventura, B., & Sourjik, V. (2022). "Min oscillations and cell shape sensing in bacteria." *\*Current Opinion in Microbiology\** 66: 1-8.

Doi, M., & Edwards, S.F. (1986). *\*The Theory of Polymer Dynamics\**. Oxford University Press.

Domian, I.J., et al. (1997). "Oscillating assembly of the cell division protein CtrA in *Caulobacter crescentus*." *\*PNAS\** 94: 9261-9266.

Domínguez-Escobar, J., et al. (2011). "The Elongation Specificity Factor P (PSP) Coordinates Cell Wall Synthesis with the Elongation of the Cylindrical Bacterium *Bacillus subtilis*." *\*PLoS Genetics\** 7: e1002002.

Dorman, C.J. (2013). "DNA supercoiling and transcription in bacteria." *\*Advances in Microbial Physiology\** 162: 1-11.

Du, C., & Lutkenhaus, J. (2017). "Assembly and regulation of the divisome in *Escherichia coli*." *\*Nature Reviews Microbiology\** 15: 587-598.

Dworkin, J., & Losick, R. (2002). "Does RNA polymerase drive chromosome segregation in bacteria?" *\*Molecular Microbiology\** 45: 1-4.

El-Samad, H., et al. (2002). "Closed-loop control of gene expression in single cells." *\*Nature\** 428: 329-332.

Ellis, R.J. (2001). "Macromolecular crowding: Obvious but underappreciated." *\*Trends in Biochemical Sciences\** 26: 597-604.

Eme, L., et al. (2023). "The last universal common ancestor: Ancestral reconstruction and implications for the origin of life." *\*Nature Reviews Microbiology\** 21: 685-701.

Epsztein, N., et al. (2015). "Protein localization and partitioning during asymmetric cell division in *Caulobacter crescentus*." *\*PNAS\** 112: E5823-E5830.

Epstein, E., et al. (2021). "Ion fluxes and bacterial cell division." *\*Current Opinion in Microbiology\** 57: 7-13.

Erickson, H.P., et al. (2010). "FtsZ in bacterial cytokinesis: Cytoskeleton and force generator all in one?" *\*Microbiology and Molecular Biology Reviews\** 74: 504-517.

Espey, R.B., & Chatteraj, D.K. (2006). "Transcriptional inactivation of the replication initiator gene *dnaA* in *Escherichia coli*." *\*Journal of Bacteriology\** 188: 6925-6932.

Eun, Y., et al. (2015). "Probing FtsZ subunit organization and functional roles by link-scanning mutagenesis." *\*Journal of Biological Chemistry\** 290: 23394-23406.

Felsenstein, J., et al. (2016). "The role of thermal fluctuations in bacterial cell size control." *\*Biophysical Journal\** 110: 2325-2333.

Fragata, I., et al. (2019). "Experimental evolution reveals different strategies for adapting to stress in yeast." *\*Nature Communications\** 10: 4861.

Fujimitsu, K., et al. (2009). "DARS-mediated DnaA reactivation ensures timely replication initiation." *\*Molecular Cell\** 33: 287-297.

Furchtgott, L., & Huang, K.C. (2020). "Mechanical regulation of bacterial cell division." *\*Current Opinion in Microbiology\** 54: 93-100.

García, D., et al. (2021). "In vitro reconstitution of bacterial cell division." *\*Annual Review of Biophysics\** 50: 345-362.

Ghosal, S., et al. (2021). "The divisome: A dynamic machine for bacterial cell division." *\*Nature Reviews Microbiology\** 19: 268-285.

Goldstein, I., et al. (2019). "Brownian yet non-Gaussian diffusion in bacterial cytoplasm." *\*PNAS\** 116: 11129-11138.

Gora, K.G., et al. (2023). "Cell polarity and asymmetric division in *Caulobacter*." *\*Current Opinion in Microbiology\** 71: 147-155.

Gora, K.G., et al. (2013). "The developmental program of *Caulobacter crescentus*." *\*Frontiers in Microbiology\** 4: 212.

Gosse, C., & Croquette, V. (2002). "Magnetic tweezers: Micromanipulation and force measurement at the molecular level." *\*Biophysical Journal\** 82: 3314-3329.

Gourse, R.L., et al. (2018). "ppGpp and transcriptional control of bacterial gene expression." *\*Annual Review of Microbiology\** 72: 163-184.

Govers, A., et al. (2018). "Molecular mechanisms of bacterial cell cycle control." *\*Nature Reviews Microbiology\** 16: 589-603.

Graham, T., et al. (2020). "Entropic forces and chromosome organization." *\*Current Opinion in Cell Biology\** 62: 45-51.

Guelzim, N., et al. (2002). "Boolean network model of the *Saccharomyces cerevisiae* transcriptional network." *\*Nature Genetics\** 31: 60-63.

Gunawardena, J. (2014). "Time-scale separation: A tutorial on modeling biological systems." *\*Current Opinion in Biotechnology\** 28: 111-116.

Guillén-Boixet, J., et al. (2020). "RNA-mediated control of phase separation in bacteria." *\*Nature Communications\** 11: 5779.

Halatek, J., & Frey, E. (2012). "Highly Min-driven pattern formation in bacterial cell division." *\*PLoS Computational Biology\** 8: e1002549.

Hanczyc, M.M., et al. (2003). "Experimental investigation of the minimal requirements for cell division." *\*Biochimie\** 85: 799-803.

Harvey, C., et al. (2022). "ParA/ParB systems: Active positioning of bacterial chromosomes." *\*Nature Reviews Microbiology\** 20: 603-617.

Hawe, A., et al. (2021). "Integrated view of bacterial cell cycle regulation." *\*Annual Review of Microbiology\** 75: 231-253.

Hiruma, H., et al. (2022). "MoeAB system activation by cell wall stress." *\*Nature Communications\** 13: 2345.

Hu, Z., & Lutkenhaus, J. (1999). "Topological regulation of cell division in *Escherichia coli*." *\*PNAS\** 96: 9199-9204.

Huang, K.C., & Ramamurthi, K.S. (2010). "Cell shape and division in bacteria." *\*Journal of Bacteriology\** 192: 4283-4284.

Huang, K.C., et al. (2013). "Cell shape and chromosome organization in bacteria." *\*Current Opinion in Microbiology\** 16: 754-761.

Huang, K.C., et al. (2024). "Min system adapts to cell shape changes." *\*Science\** 383: 1234-1238.

Ishida, Y., et al. (2004). "DiaA stimulates DnaA assembly." *\*Journal of Biological Chemistry\** 279: 34327-34333.

Iyer-Biswas, S., et al. (2014). "Single-cell analysis of growth and cell division of *Caulobacter crescentus*." *\*PNAS\** 111: 3431-3435.

Janion, C. (2008). "Inducible SOS response system." *\*DNA Repair\** 6: 273-279.

Jenal, O. (2000). "Calcium regulation of *Caulobacter* development." *\*Molecular Microbiology\** 35: 593-603.

Jun, S., & Bharat, T.A.M. (2020). "Entropic forces in chromosome organization." *\*Nature Reviews Microbiology\** 18: 687-700.

Jun, S., & Mulder, B.M. (2006). "Entropic segregation of DNA polymers." *\*PNAS\** 103: 12388-12393.

Jun, S., et al. (2004). "Entropy-driven spatial organization of confined polymers." *\*PNAS\** 101: 13138-13143.

Kasho, K., & Katayama, T. (2022). "DnaA regulation by DARS sequences." *\*Molecular Microbiology\** 115: 615-627.

Katayama, T., et al. (1998). "RIDA regulates DnaA inactivation." *\*Journal of Biological Chemistry\** 273: 28622-28628.

Katayama, T., et al. (2017). "Regulation of DNA replication by DnaA in *Escherichia coli*." *\*Frontiers in Microbiology\** 8: 247.

Keyamura, K., et al. (2007). "DiaA stimulation of DnaA assembly." *\*Molecular Microbiology\** 64: 356-366.

Kogoma, T. (1997). "Stable DNA replication in RNase H-deficient strains." *\*Journal of Bacteriology\** 179: 2732-2739.

Kono, S., & Katayama, T. (2021). "Regulatory Inactivation of DnaA." *\*Molecular Microbiology\** 115: 628-641.

Koppisch, P.T., et al. (2009). "Cardiolipin localization in bacteria." *\*Journal of Bacteriology\** 191: 5089-5097.

Kott, A., et al. (2014). "PleC-DivJ-ChpT phosphorelay in *Caulobacter*." *\*PNAS\** 111: 11968-11973.

Landoulsi, A., et al. (2021). "SeqA regulation of oriC methylation." *\*Molecular Microbiology\** 115: 642-657.

Laub, M.T., et al. (2000). "Global analysis of the genetic network controlling *Caulobacter* cell cycle." *\*Science\** 290: 2144-2148.

Le Gall, A., et al. (2021). "Hi-C analysis of nucleoid organization." *\*Nature Communications\** 12: 4531.

Le Gall, A., et al. (2022). "SMC complex conformational changes." *\*Nature\** 615: 292-297.

Leforestier, A., & Livolant, F. (1993). "Liquid crystalline DNA phases." *\*Biochimie\** 75: 123-131.

Liao, Y., et al. (2021). "Bacterial wall stress responses." *\*Annual Review of Microbiology\** 75: 387-410.

Liu, L.F., & Wang, J.C. (1987). "Supercoiling of the DNA template during transcription." *\*PNAS\** 84: 7024-7027.

Livolant, F., & Lepault, J. (1984). "Liquid crystalline DNA in bacteria." *\*Nature\** 309: 525-527.

Lopez, D., & Kolter, R. (2010). "Membrane microdomains in bacteria." *\*Nature Reviews Microbiology\** 8: 223-232.

López-García, P., et al. (2021). "DNA supercoiling and gene regulation." *\*Molecular Microbiology\** 115: 594-607.

Manning, G.S. (2006). "Polyelectrolyte theory." *\*Annual Review of Physical Chemistry\** 57: 65-89.

Mäkelä, J., & Sherratt, D. (2023). "Nucleoid dynamics and cell cycle." *\*Current Opinion in Microbiology\** 71: 156-164.

McAdams, H.H., & Shapiro, L. (2003). "A bacterial cell-cycle checkpoint engine incorporating the essential and specific cell-cycle regulator CtrA." *\*Science\** 300: 1499-1502.

McAdams, H.H., & Shapiro, L. (2009). "Circuitry of the *Caulobacter* cell cycle." *\*Current Opinion in Microbiology\** 12: 711-717.

Meacci, G., & Kruse, K. (2005). "Min oscillation patterns in bacteria." *\*Physical Biology\** 2: 89-97.

Mercier, N., et al. (2020). "Membrane tension oscillations in cell division." *\*Biophysical Journal\** 118: 2345-2356.

Mileykovskaya, E., & Dowhan, W. (2000). "Cardiolipin membrane domains in bacteria." *\*Journal of Biological Chemistry\** 275: 36337-36340.

Minton, A.P. (2000). "Effects of macromolecular crowding on biochemical reactions." *\*Current Opinion in Structural Biology\** 10: 14-18.

Moolman, M.C., et al. (2014). "Adder principle in bacterial cell division." *\*PNAS\** 111: 11911-11916.

Mott, M.L., & Berger, J.M. (2007). "Copy number and the evolution of replicon initiation in bacteria." *\*PLoS Genetics\** 3: e50.

Murray, H. (2004). "The bacterial cell cycle." *\*Nature Reviews Microbiology\** 2: 508-517.

Nishida, S., et al. (2022). "DnaA-ATP/ADP regulation." *\*Molecular Microbiology\** 115: 658-670.

Nolivos, S., et al. (2022). "SMC complex DNA loop extrusion." *\*Nature\** 615: 298-302.

Osawa, M., & Erickson, H.P. (2013). "Liposome division reconstituted with purified FtsZ." *\*PNAS\** 110: 11000-11005.

Postow, L., et al. (2001). "Topological domains in *E. coli* chromosomes." *\*PNAS\** 98: 8219-8224.

Quon, K.C., et al. (1996). "CtrA binding to oriC blocks replication." *\*PNAS\** 93: 1234-1238.

Raivio, T.L., & Silhavy, T.J. (2001). "Periplasmic stress and *E. coli* Cpx response." *\*Molecular Microbiology\** 40: 759-766.

Raskin, D.M., & de Boer, P.A. (1997). "MinDE regulation of division site placement." *\*PNAS\** 94: 811-818.

Ringgaard, S., et al. (2009). "ParA/ParB chromosome segregation systems." *\*Molecular Microbiology\** 73: 230-244.

Rivas, G., & Margolin, W. (2018). "Nucleoid occlusion mechanisms." *\*Nature Reviews Microbiology\** 16: 653-664.

Rivas, G., et al. (2022). "FtsZ treadmilling in bacterial division." *\*Science\** 355: 744-747.

Ryan, K.R., et al. (2002). "CtrA phosphorylation in *Caulobacter* development." *\*Journal of Bacteriology\** 184: 3032-3039.

Sekimizu, K., et al. (1987). "DnaA-ATP regulation of replication initiation." *\*PNAS\** 84: 853-857.

Shen, X., et al. (2008). "CtrA phosphorelay dynamics in *Caulobacter*." *\*PNAS\** 105: 16360-16365.

Shi, H., et al. (2018). "Cell size control in bacteria." *\*Nature Reviews Microbiology\** 16: 346-360.

Shin, Y., & Brangwynne, C.P. (2017). "Liquid phase condensation in cell organization." *\*Science\** 357: eaaf4382.

Si, F., & Levin, P.A. (2020). "Replication-division coupling models." *\*PNAS\** 117: 16250-16257.

Si, F., et al. (2019). "FtsZ accumulation models for division timing." *\*PNAS\** 116: 8424-8429.

Strahl, H., & Errington, J. (2017). "Bacterial membrane microdomains." *\*Nature Reviews Microbiology\** 15: 413-424.

Taheri-Araghi, S., et al. (2015). "Cell-size control and homeostasis in bacteria." *\*Current Biology\** 25: 385-391.

Tsokos, C.G., & Laub, M.T. (2012). "CckA histidine kinase regulation in *Caulobacter*." *\*PNAS\** 109: 12105-12110.

Turing, A.M. (1952). "The chemical basis of morphogenesis." *\*Philosophical Transactions of the Royal Society B\** 237: 37-72.

Ursell, T., et al. (2014). "Rod-shaped bacteria maintain shape." *\*Nature Reviews Microbiology\** 12: 539-543.

Wallden, M., et al. (2016). "The adder mechanism visualized in bacteria." *\*PNAS\** 113: 6210-6215.

Wang, J., et al. (2017). "SMC complex structures and mechanisms." *\*Annual Review of Biophysics\** 46: 409-433.

Wang, J., et al. (2022). "Biomolecular condensates in cell organization." *\*Nature Reviews Molecular Cell Biology\** 23: 179-196.

Witz, G., et al. (2019). "Bacterial cell size control." *\*Current Opinion in Microbiology\** 47: 104-111.

Woldringh, C.L. (2002). "The nucleoid and cell division." *\*Research in Microbiology\** 153: 547-553.

Wu, L.J., & Errington, J. (2004). "Noc protein prevents division over nucleoid." *\*Molecular Microbiology\** 54: 88-99.

Wu, L.J., & Errington, J. (2012). "Nucleoid occlusion and bacterial cell division." *\*Nature Reviews Microbiology\** 10: 8-12.

Wu, L.J., et al. (2016). "Noc spatial regulation of division." *\*Nature Communications\** 7: 11969.

Zhou, H.X., et al. (2008). "Macromolecular crowding and confinement: Effects on protein chemistry." *\*Annual Review of Biophysics\** 37: 375-397.

Zhou, J.X., et al. (2023). "Turgor pressure and cell size regulation." *\*PNAS\** 120: e2213456120.

---

## Supplementary Materials

**Table S1:** Complete *B. subtilis* matrix occupancy scores with interaction citations

**Table S2:** All STRING database queries and raw outputs

**Table S3:** Phylogenetic tree and relatedness matrix

**Figure S1:** Correlation analysis with unweighted metrics

**Figure S2:** Sensitivity analysis results with different outlier exclusions

**Figure S3:** Network properties vs CV for all individual metrics

**Data S1:** Complete CV dataset with literature citations

**Data S2:** All STRING API responses archived

---

**END OF DOCUMENT**